



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Mid Term Exam:: Trimester: Summer 2022

Course Code: CSE 1111, Course Title: Structured Programming Language

Total Marks: 30

Duration: 1:45 hour

There are FIVE questions. Answer all the questions. Marks are indicated in the right margin.

Q.1 a) Rewrite the following code after correcting the errors. [2]

```
include <stdio.h>
void main() {
    int num1 = 5, float num2, char chr = 'q';
    scanf("%d", num2);
    num1 = num2 % chr;
    printf("Result is = %f ", num1);
}
```

b) Identify the invalid variable names from the following. Mention the reasons that make them invalid. [2]

largest_val, smallest-val, while, 2ndNum, !New, avg marks, val9

c) Compute the values of the variables a, b, c, and d. [2]

float a=5*(5/2), int b=5*(5/2), float c=5*(5.0/2), int d=5*(5.0/2)

Q.2 a) Write down the output of the following C program, for num = 1 and num = 3. [3]

```
#include <stdio.h>
int main() {
    int num;
    int sum = 0, i = 10, j = 5;
    scanf("%d", &num);
    switch(num) {
        case 1:
            sum = 2*i++;
            j++;
        case 2:
            sum = 2*j--;
            i++;
            break;
        case 3:
            sum = ++i*j--;
        case 4:
            sum = i++*j--;
        default:
            sum=0;
            i=0;
            j=0;
    }
    printf("%d %d %d", i, j, sum);
    return 0;
}
```

b) Manually trace the following code segment and show the change of values of the variables i, sum, b, a, y, x in each step. [3]

```
int sum=0, i, a = 1, b, x = 1, y = 1;
for(i=1; i<=5; i++) {
    sum = sum + a;
    b = 6*x + 1;
    a = a + b;
    y++;
    x = x + y;
}
```

- Q.3 a)** Replace the nested “for” loop in the following code using nested “do-while” loop **without** [3]
changing the logical meaning of the program:

```
void main() {
    int n = 3, i, j, sum = 0;
    for(i = 0; i < n; i++) {
        for(j = 0; j <= i; j++) {
            if(i == j) sum += i + j;
            else if(i > j) sum += i + n;
            else sum += n - j;
        }
    }
}
```

- b) Write a program to find the **online average** of the **positive numbers** given as inputs by the [3]
user. To solve this problem, you should do the following:
- Write an **infinite loop** that will terminate if the user gives 0 as input.
 - If the user gives a **positive number** as input, you should keep adding it.
 - You should also **keep track** of how many positive numbers are given as inputs.
 - Finally, when the loop terminates, you should **calculate the average** by dividing the sum of the positive numbers by the total positive numbers.

- Q.4 a)** Show the **manual tracing** (show the values of all the variables and array elements in each [3]
step) for the following code segment.

```
int F[6] = {0};
int i, n;
n = 3;
for(i = 0; i < 6; i++){
    F[i] = n+i;
    if(F[i]%2 == 0){
        F[i] *= 2;
    }
}
```

- b) Write a program to perform the following operation: [3]
- Read **n integer numbers** from keyboard and **store** them in an array of size 100, where n is input integer from keyboard.
 - Print** all the array elements with their indices (plural of index) in the following format.

Index	Value
0	11
1	7

- Find and print the **average** of the numbers that are stored in **odd numbered indices** in the array.

- Q.5 a)** Draw a **flowchart** to find the **sum** of the following series up to n terms, where n is input integer [3]
number from keyboard.

$1 - 2 + 3 - 4 + \dots \text{upto } n \text{ terms}$

- b) Write a program that takes an integer **n** as input from the user and **prints** the following [3]
pattern. Program for n, NOT 3 or 5.

Sample input, n	Sample output
3	6 4 2 4 2 2
5	10 8 6 4 2 8 6 4 2 6 4 2 4 2 2